Predicting Credit Card Defaulting

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Credit Card Default Prediction Project

NYU Tandon Machine Learning Final Project

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11:00 A.M Section

CS-UY 4563

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Introduction

Dataset contains 1250 credit card clients with information on:

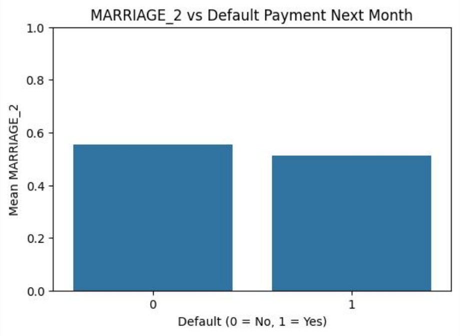
1. default payments
2. demographic factors
3. credit data
4. history of payment
5. bill statements

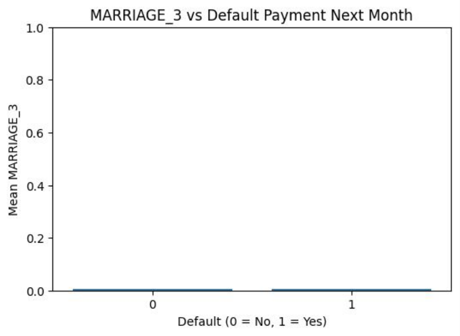
After an initial exploration of the dataset:

1. There are 24 features and 1 target variable
2. The dataset contains 959 clients who did not default the following month and 291 who did

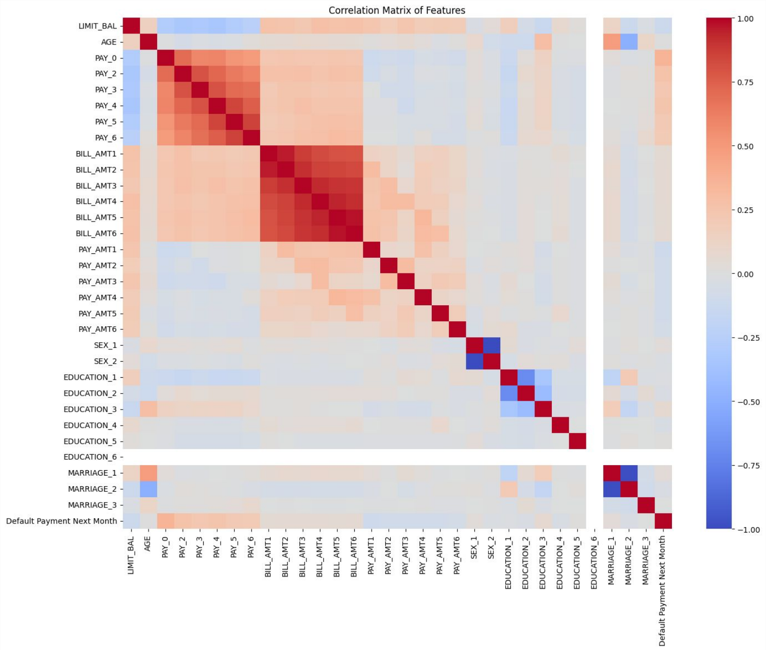
Preprocessing

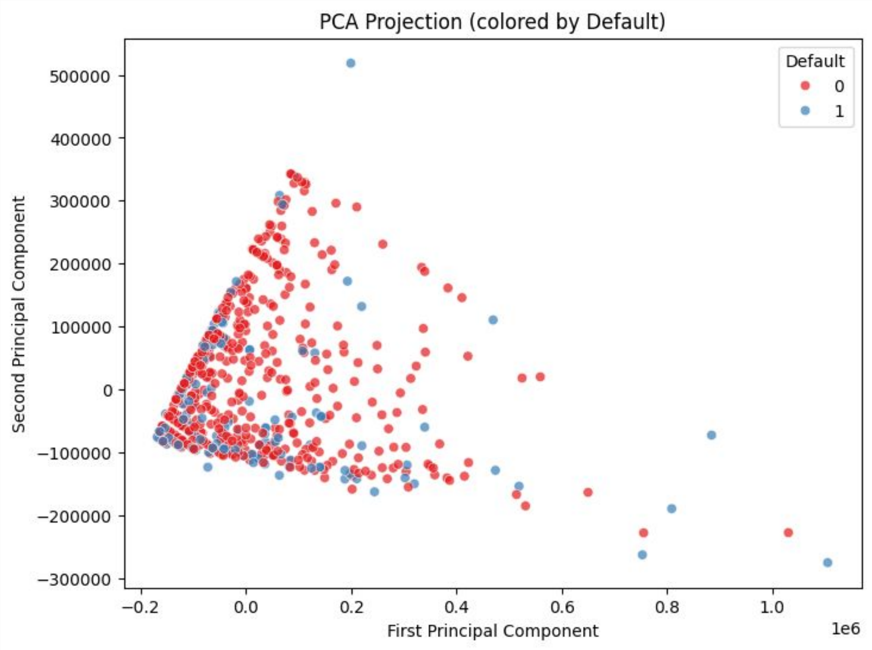
1. Removed incomplete values
2. Applied StandardScaler()
3. One-Hot Encoded the categorical variable
4. Used feature reduction to combine similar features and remove sparse variables





Important Visualizations





Transformations and Hyperparameters

Feature Transformations:

- None (linear)
- Polynomial (2, 3)
- RBF
- PCA

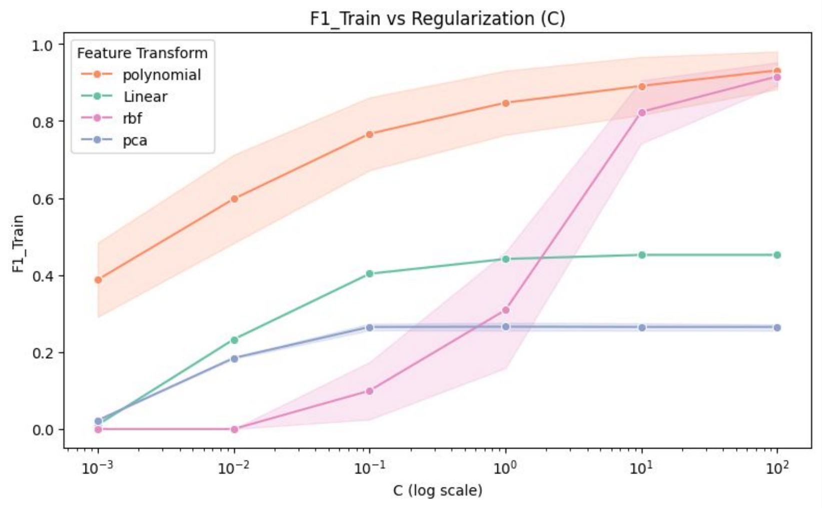
Hyperparameters:

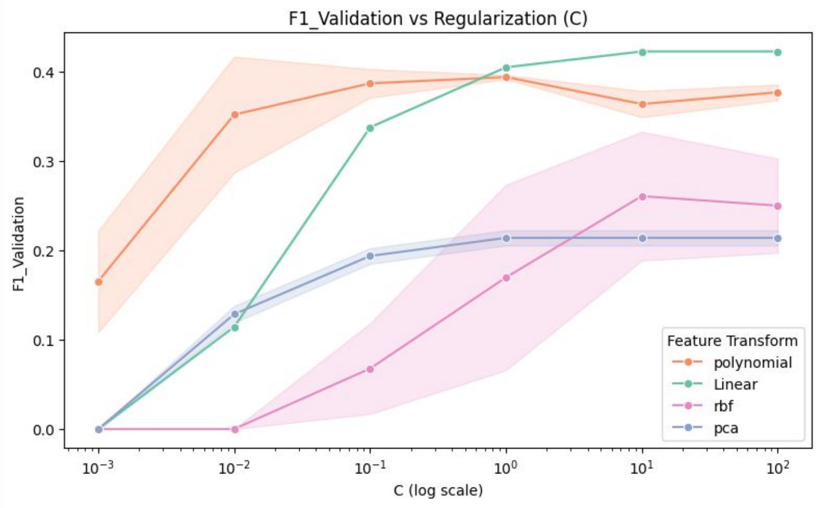
- Gamma (RBF)
- C = [0.001,0.01, 0.1, 1, 10, 100]

Neural Network Hyperparameters:

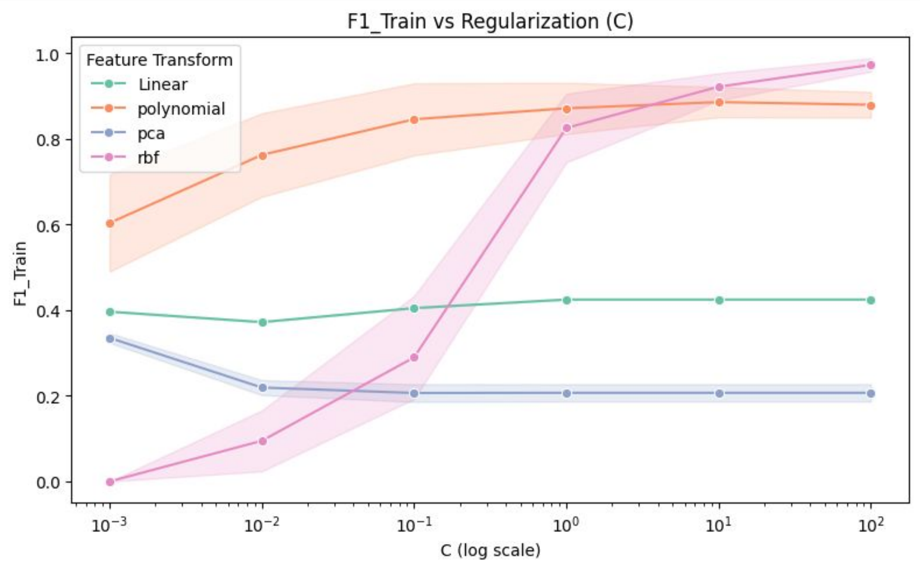
- Hidden Layer Structure
- Alpha (L2 weight decay)
- Initial Learning Rate

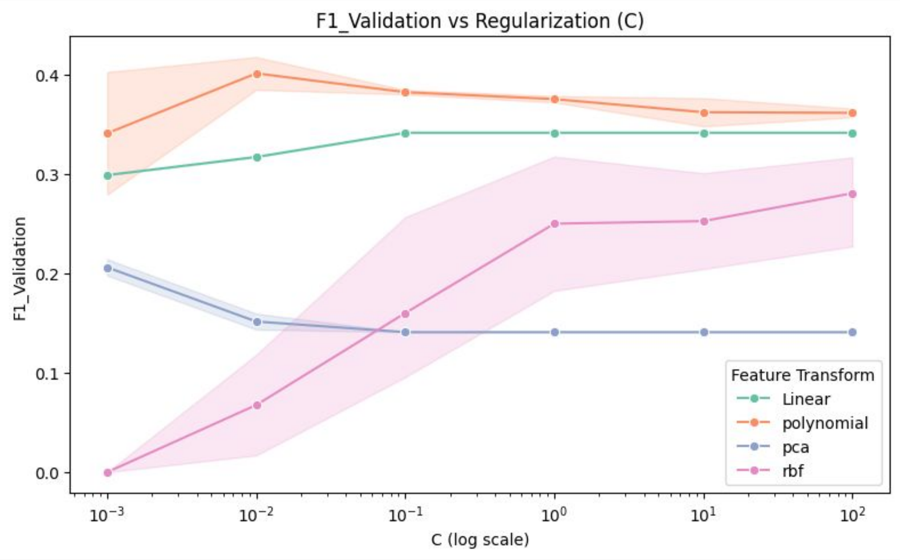
Logistic Regression



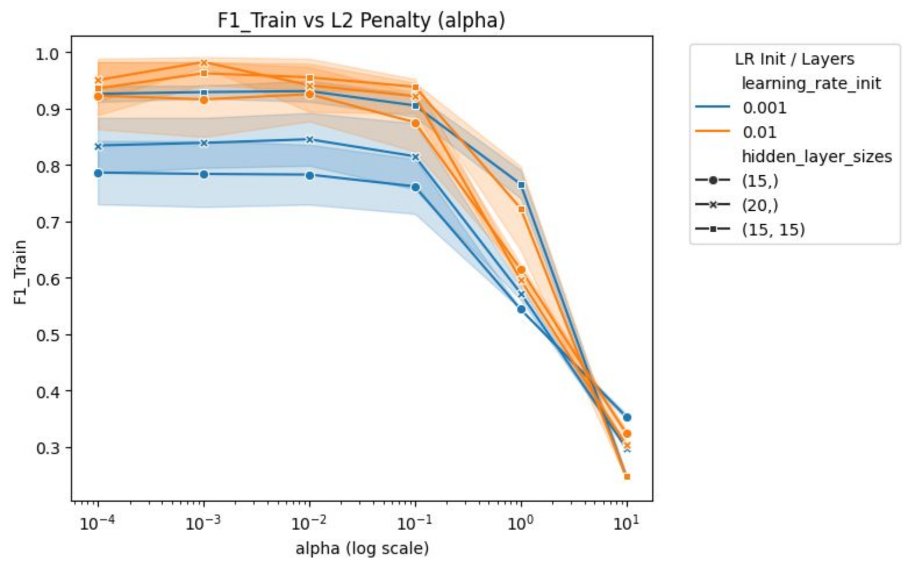


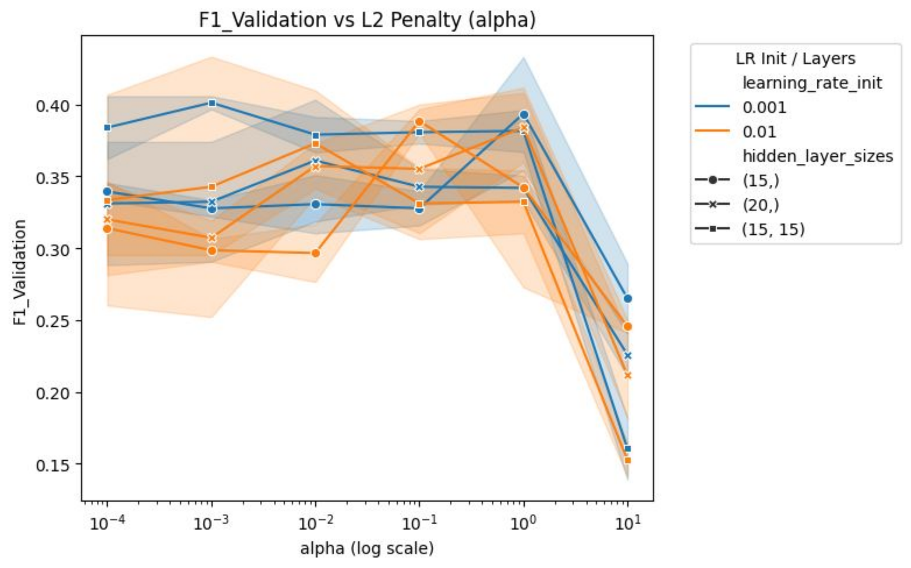
SVM


F1_Train vs Regularization (C)
Feature Transform
Linear
polynomial
pca
rbf
F1_Train
1.0
0.8
0.6
0.4
0.2
0.0
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
C (log scale)


F1_Validation vs Regularization (C)
F1_Validation
0.4
0.3
0.2
0.1
0.0
Feature Transform
Linear
polynomial
pca
rbf
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
C (log scale)

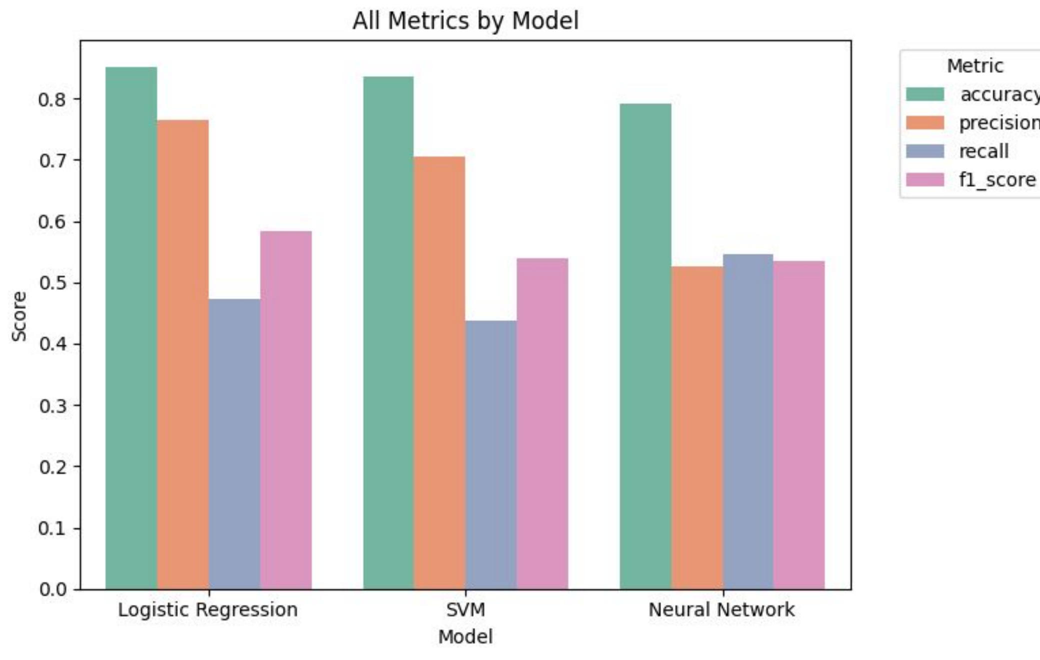
Neural Networks





Results

1. Models seemed to suffer from overfitting despite regularization
2. PCA and RBF were not effective
3. Polynomial feature transformation on logistic regression performed well



Improvements

- Better feature engineering
- Resampling
- Experimenting with more hyperparameters